

SHREYAS RAORANE

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Summary

Robotics Software Engineer working at the boundary of autonomy software and the systems it runs on. M.S.E. candidate at Penn's GRASP Lab. Built time-critical, multi-threaded C++ systems with millisecond-scale timing at OPEX Corporation, and a full ROS2 autonomy stack processing LIDAR, IMU, and RGB-D sensor data on Linux.

Education

University of Pennsylvania

Philadelphia, PA

Master of Science in Engineering (M.S.E.), Robotics GPA: 3.80/4.0 (Summa Cum laude)

Aug 2024 – Jun 2026

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology GPA: 3.91/4.0 (First Class with Distinction)

Jun 2018 – Jun 2022

Skills

Programming	C++, Python, MATLAB, Object-Oriented Design, Design Patterns
Systems	Multi-threading, Real-time Systems, Memory Management, Event-driven Architecture, Linux
Robotics	ROS2, Sensor Integration (LIDAR, IMU, RGB-D), Localization, Multi-Agent Coordination
Motion Planning	RRT*, A*, MPC, DMPC, QP Optimization, Sampling-Based Planning, Collision Avoidance
Tools & Sim	Gazebo, PyBullet, AutoDRIVE, Foxglove, Docker, Git, CI/CD

Work Experience

EvoSort Simulation

Jun – Aug 2025

OPEX Corporation, Time Critical Software Engineering Intern (3 Months)

Moorestown, NJ

- Built multi-threaded discrete-event simulation in C++ for warehouse automation, managing 1,000+ parallel material flow events with shared state across 3 concurrent threads
- Developed time-critical scheduling engine completing actuation sequences within a few milliseconds, modeling real hardware timing for production validation
- Refactored legacy codebase with shared header architecture and a centralized `SimulationHandler`, letting the software team validate against the simulator instead of waiting for limited physical hardware availability

ESAP Migration Project

Jun 2022 – Jun 2024

Brillio Technologies, Software Development Engineer (2 Years)

Bengaluru, India

- Delivered production software for Verizon's ESAP Migration Project (10,000+ users); optimized 40+ PostgreSQL queries reducing response time by 35%

Robotics Projects

F1/10 Autonomous Racing Stack

Feb – Mar 2025

University of Pennsylvania

- Built full ROS2 autonomy stack on Linux integrating LIDAR, IMU, and RGB-D sensor data for perception, localization, and planning; bridged the AutoDRIVE Simulator and F1/10 hardware over a shared sensor and message interface, with a live tuning interface replacing the recompile-and-redeploy loop
- Repository: github.com/Shreyas0812/roboracer-ws

SBAMP: Sampling-Based Adaptive Motion Planning

Mar – Apr 2025

University of Pennsylvania

- Built adaptive RRT* planner integrated with dynamical systems attractors for real-time goal tracking on mobile robots; replanned fast enough to track goals at speeds where the static RRT* baseline could not keep up
- Repository: github.com/Shreyas0812/SBAMP

SwarmSync: Distributed MPC for Quadrotor Swarms

Oct – Dec 2025

University of Pennsylvania

- Implemented DMPC for collision-free trajectory generation across 8+ quadrotors using QP optimization; agents reliably reached goals in simulation with dynamic obstacles, degrading only in extremely dense environments
- Repository: github.com/Shreyas0812/SwarmSync

Research Experience

Master's Thesis: Scalable Multi-Robot Coordination

Oct 2025 – May 2026

University of Pennsylvania, GRASP Lab Advisors: Prof. Linh Phan, Prof. Rahul Mangharam, Prof. M. Ani Hsieh

LCBA Task Allocation with Decentralized Path Planning

- Designed LCBA (Local Consensus Bundle Algorithm), a decentralized task allocation algorithm extending GCBBA to intermittently disconnected, dynamic communication graphs via component-local consensus and event-driven replanning; maintains non-zero task completion where CBBA and SGA baselines drop to zero
 - Integrated LCBA with three interchangeable path planners (Cooperative A*, PBS, RHCR); evaluated across six maps from 6 to 200 agents over 432+ runs against CBBA, SGA, and DMCHBA, achieving a 0.91 makespan ratio vs. centralized SGA
- Repository: github.com/Shreyas0812/GCBBA-Warehouse-System